

Multimedia Computing

Course Title: Multimedia Computing
Course No: BIT356
Nature of the Course: Theory + Lab
Semester: VI

Full Marks: 60 + 20 + 20
Pass Marks: 24 + 8 + 8
Credit Hrs: 3

Course Description: This course familiarizes students with the concepts of multimedia computing including sound, image, video, animations, data compression, and multimedia applications.

Course Objectives: The main objective of this course is to provide knowledge of different concepts of multimedia computing and their applications.

Course Contents:

Unit 1: Introduction (2 Hrs.)

Multimedia Systems; Components of multimedia; Multimedia System, its characteristics and properties; Structure of Multimedia; Applications

Unit 2: Text / Sound / Audio System (7 Hrs.)

Text: Types of Text, Ways to Present Text, Aspects of Text Design, Character, Character Set, Codes, Unicode, Encryption

Sound / Audio: Basic Sound Concepts, Types of Sound, Digitizing Sound, Audio Formats, Audio tools, MIDI; Speech Generation, Speech Analysis and Speech Transmission

Unit 3: Image and Video (6 Hrs.)

Image: Digital Image Representation, Uses of Images, Formats, Image Color Scheme, Image Enhancement; Image Synthesis, Analysis and Transmission

Video: Analogue and Digital Video, Recording Formats and Standards (JPEG, MPEG, H.261) Transmission of Video Signals, Video Capture

Unit 4: Video and Animation (8 Hrs.)

Digital Video; Video Signal Representation; Computer Video Format; Computer-Based Animation; Animation Language; Timeline and Frame based animation; Timeline and Tween-Based Animation; Methods of controlling Animation; Display of Animation; Transmission of Animation

Unit 5: Data Compression (5 Hrs.)

Introduction; basics of compression; storage space; need of coding; Lossy and Lossless Compression Techniques; Source, Entropy and Hybrid Coding; Lossy Sequential DCT – based mode; Expanded Lossy DCT-based mode; JPEG and MPEG Compression

Unit 6: Abstraction for Programming (5 Hrs.)

Introduction to abstraction levels; Libraries; System Software; Toolkits; Higher Programming Languages; Object Oriented Approaches

Unit 7: Multimedia Design (4 Hrs.)

Development Phases and Development Teams; Analysis, Design, Development and Implementation Phase; Evaluation and Testing Phase

Unit 8: User Interfaces (4 Hrs.)

Basic Design Issues; Video and Audio at the User Interface; User-Friendliness as the Primary Goal

Unit 9: Multimedia Application (4 Hrs.)

Media Preparation and Composition; Media Integration and Communication; Media Entertainment; Telemedicine; E-learning; Digital Video Editing and Production Systems; Video Conferencing; Video-on-demand

Laboratory Works:

The laboratory works should focus on writing programs of different concepts of multimedia computing

References:

1. Multimedia: Computing, Communications and Applications, Ralf Steinmetz and Klara Nahrstedt, Pearson Education Asia
2. Multimedia Communications, Applications, Networks, Protocols and Standards, Fred Halsall, Pearson Education Asia
3. Multimedia Systems, John F. Koegel Buford, Pearson Education Asia
4. Multimedia Fundamentals: Vol. 1- Media Coding and Content Processing, Ralf Steinmetz and Klara Nahrstedt, PHI.